NHR SÜD-WEST

WELCOME AND INTRODUCTION TO NHR@SW

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Johannes Gutenberg University Mainz

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JG|U

WELCOME
Who are We?



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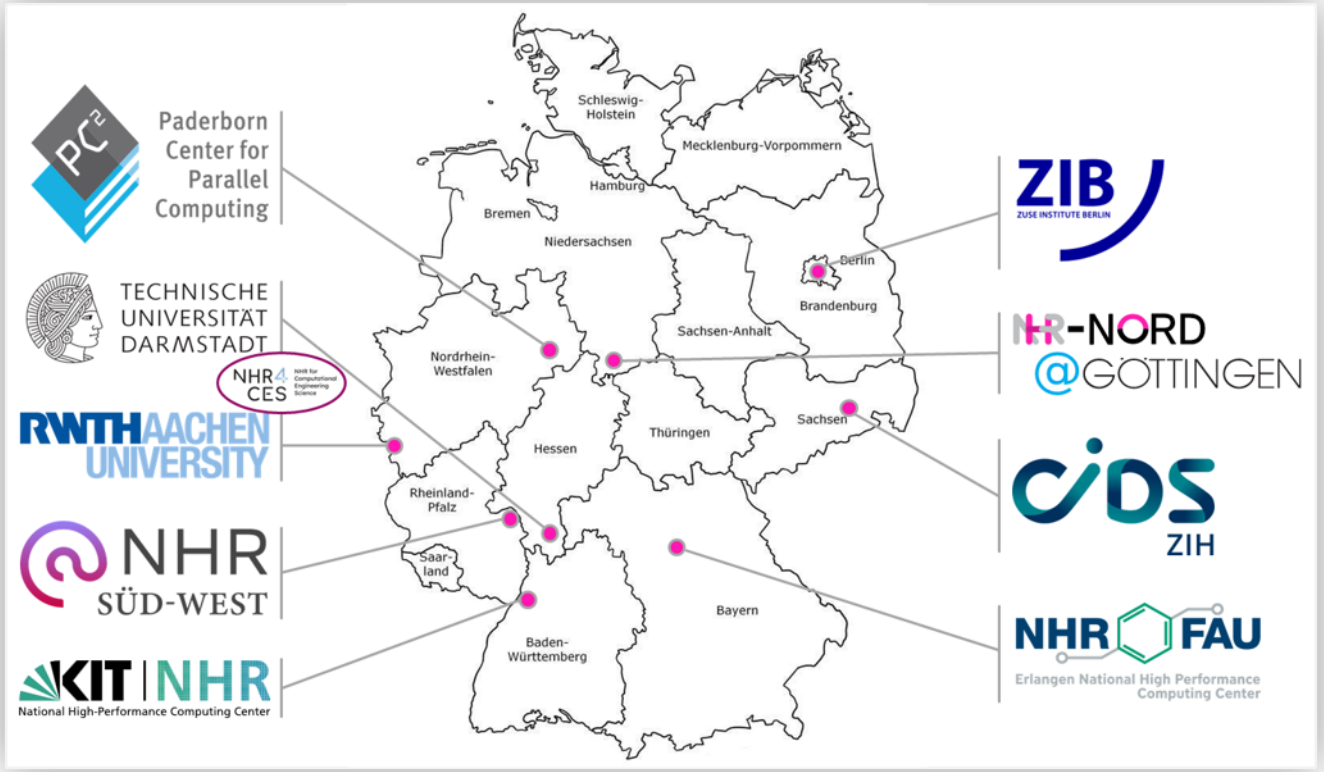
Prof. Dr. Sarah Neuwirth
neuwirth@uni-mainz.de

Introduction to NHR@SW

INTRODUCTION TO NHR@SW
What is NHR?

National High Performance Computing (NHR)

- Nationwide alliance combining HPC infrastructure and expertise at German universities
- Free access to powerful computing resources for researchers across disciplines
- Expert support and training to help researchers use HPC effectively
- Research into efficient scientific computing and support for early-career researchers through the NHR Graduate School



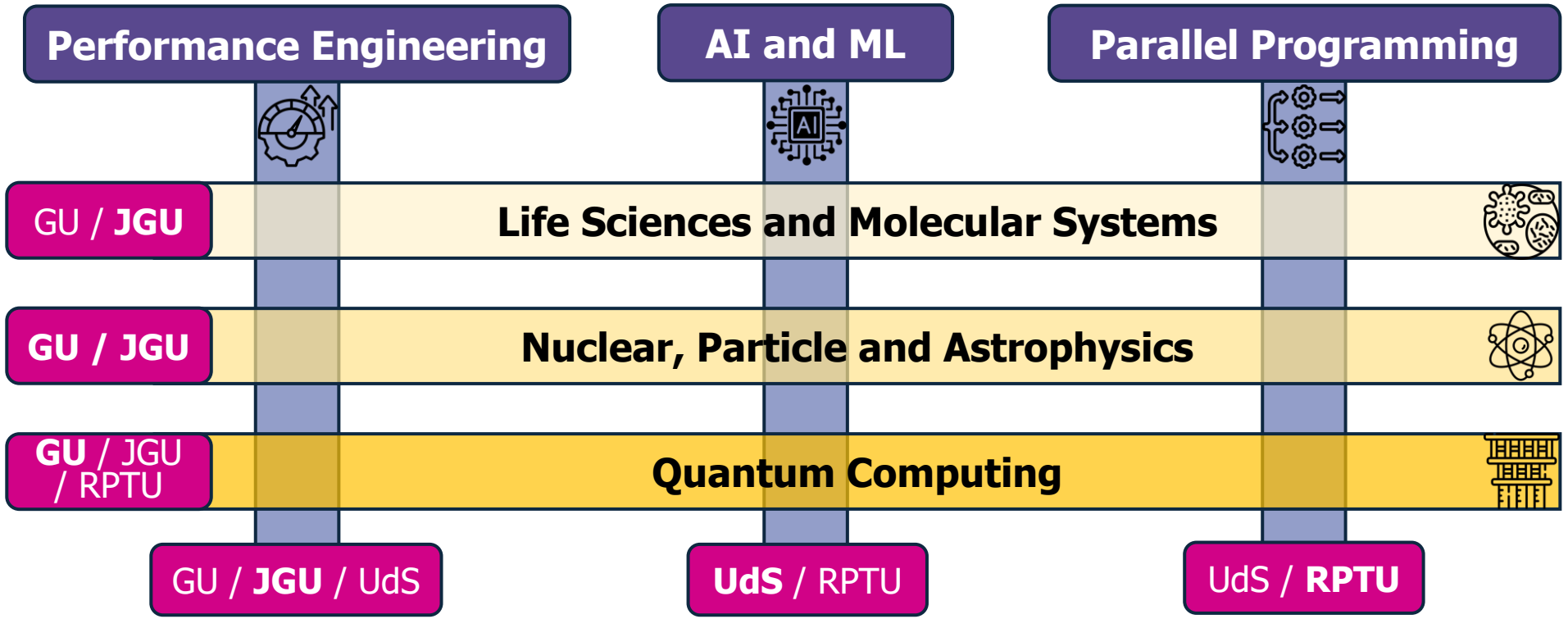
INTRODUCTION TO NHR@SW

NHR Süd-West: One Center, Four Universities

- Collaboration between Goethe University Frankfurt, JGU Mainz, RPTU Kaiserslautern-Landau and Saarland University
- Access to MOGON NHR and Goethe-NHR for researchers throughout Germany
- Individual support for scientific applications, performance optimization and data management
- Method Labs and Simulation and Data Labs connect computational methods with scientific domain expertise
- Training and community activities covering HPC, AI, parallel programming and related topics
- **Further information:**
  - Website: https://nhrsw.de/
  - Contact: office@lists.nhrsw.de



INTRODUCTION TO NHR@SW
Competence Areas and Training



Organized in two lab types: "**Method Labs**" and "**Simulation and Data Labs**" (**SDLs**)

*SDLs – **support to user domains*** *Method labs – **support both users and SDLs***

NHR@SW offers:

→ Technical expertise
→ HPC methods
→ Training programs

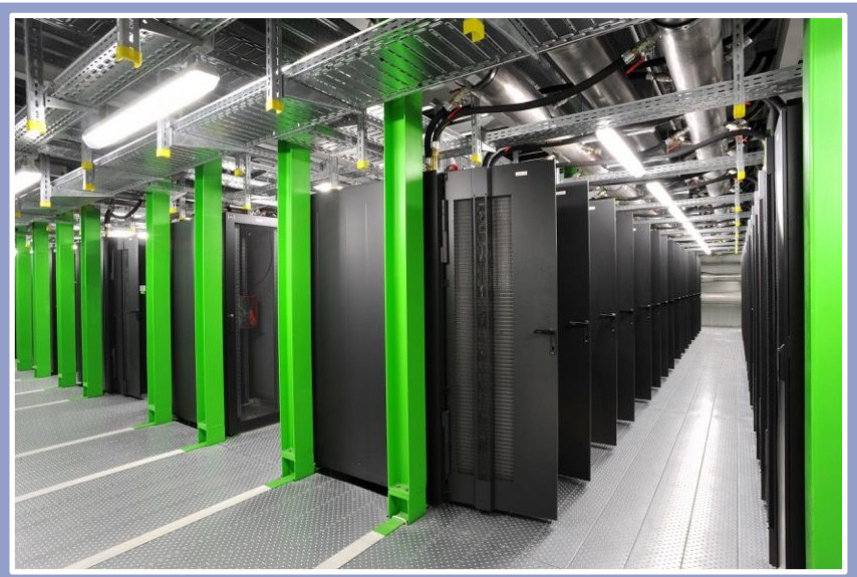
Scan me!!

INTRODUCTION TO NHR@SW
Compute Infrastructure



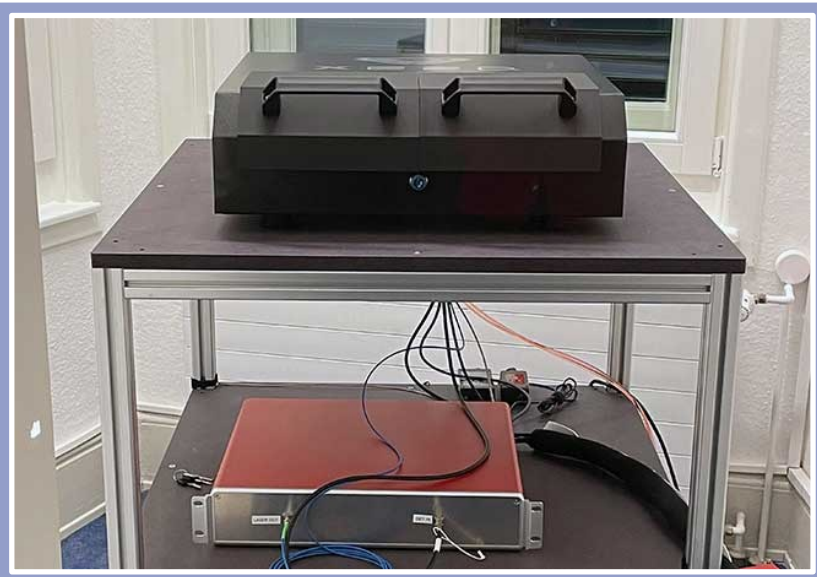
MOGON NHR

- 590 compute nodes
- 75,000 AMD EPYC 7713 cores
- 40 A100 Tensor core GPUs
- 186 TB distributed main memory
- 20 PB parallel file system
- Direct liquid cooling
- CPU R_{peak} ≈ 2.8 PF/s (DP)
- GPU R_{peak} ≈ 780 TF/s (DP)



GOETHE-NHR

- 730 compute nodes
- 27,000 CPU cores
- 864 AMD MI210 GPUs
- Up to 512 GB RAM per node
- 12 PB Parallel File System
- Air-to-liquid cooling
- CPU-Island R_{peak} ≈ 1.4 PF/s (DP)
- GPU-Island R_{peak} ≈ 9.09 PF/s (DP)



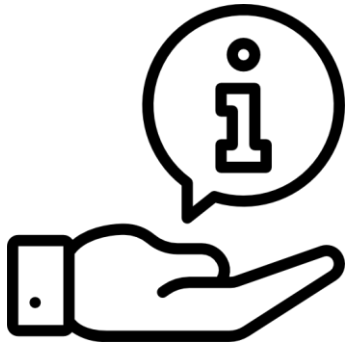
BABY DIAMOND

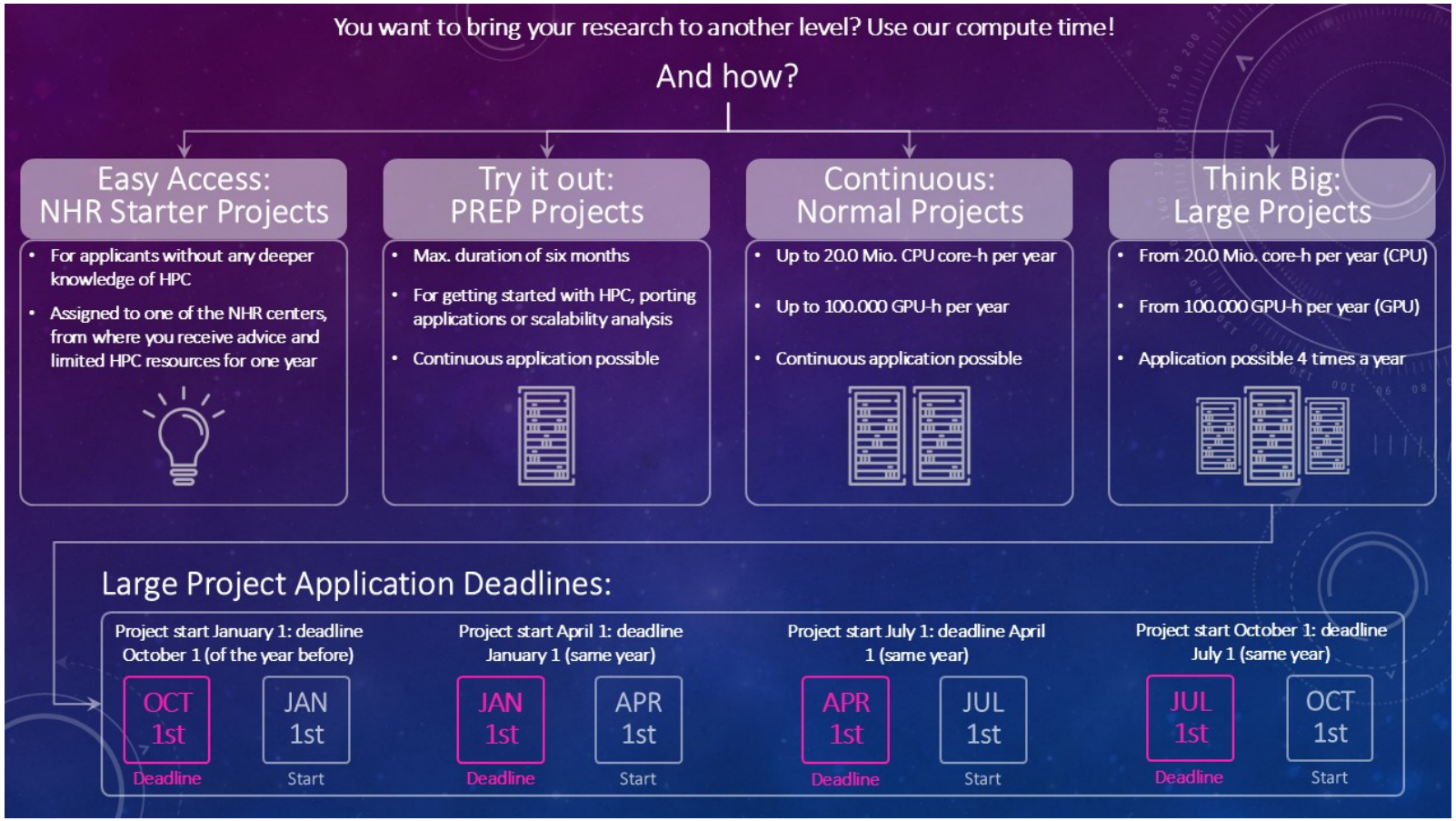
- First quantum computer at GU
- Commissioned on 16. Dec. 2024
- Nitrogen-vacancy centers in a very small artificial diamond
- Works at room temperature
- Consists of 5 qubits
- Operated via digital pulse electronics and lasers

INTRODUCTION TO NHR@SW
How to Apply for Compute Time

Compute Time Applications:
https://nhrsw.de/application/

Step-by-Step Guide:
https://nhrsw.de/hpc-access/





INTRODUCTION TO NHR@SW
Who Can Apply for Compute Time?

- PI must be a member of a German University (Universität, Fachhochschule, Musikhochschule, etc.). This does not include German research organizations like Max-Planck , Helmholtz, Frauenhofer, etc. – though many directors and group leaders there are also a member of a university and can apply in that capacity.
- PI must have a PhD or equivalent.
- These requirements do not apply to other project members like the Person of Contact (PC).
- PI can simply finalize the application online, while the PC has to upload a document signed (digital or physical) by the PI. PC can prepare the application, and the PI can log in anf finalize it to minimize work.
- **Where to apply?** https://jards.nhr-verein.de/
- **Additional Information:**
  - https://nhr-verein.de
  - https://nhrsw.de/application

NHR SÜD-WEST

Outlook on the Agenda

OUTLOOK ON THE AGENDA

Day 1: Introduction to I/O and Cluster Access

Monday, November 2, 2026	
13:30 – 13:40	Welcome and Introduction to NHR@SW
13:40 – 15:00	I/O Lecture Speaker: *Sarah Neuwirth*
15:00 – 15:30	Break
15:30 – 16:50	Mogon 101 – Introduction to Cluster Computing Speaker: *Felix Schlapp*
16:50 – 17:00	Closing Remarks / Recap Day 1

OUTLOOK ON THE AGENDA

Day 2: Darshan

Tuesday, November 3, 2026	
13:30 – 13:35	Welcome to Day 2
13:35 – 14:15	Introduction to Darshan Speaker: *Julius Athenstaedt*
14:15 – 15:00	Hands-on: Installation of Darshan & HACC-IO
15:00 – 15:30	Break
15:30 – 16:00	Hands-on: Darshan Analysis of HACC-IO
16:00 – 16:55	Hands-on: Advanced Darshan - PyDarshan & DXT
16:50 – 17:00	Closing Remarks / Recap Day 2

OUTLOOK ON THE AGENDA

Day 3: DFTracer

Wednesday, November 4, 2026	
13:30 – 13:35	Welcome to Day 3
13:35 – 14:15	Introduction to DFTracer Speaker: *Radita Liem*
14:15 – 15:00	Hands-on: installation of DFTracer & BT-IO
15:00 – 15:30	Break
15:30 – 16:00	Hands-on: Using DFAnalyzer with BT-IO
16:00 – 16:55	Hands-on: Advanced DFTracer - Annotation & ML Workloads
16:50 – 17:00	Closing Remarks / Recap Day 3

IMPORTANT INFORMATION

- Official Documentation: https://docs.hpc.uni-mainz.de/
- Cheat Sheets: https://seafile.rlp.net/d/98d53e6be8e84ca090f1/
- Connecting to compute nodes (SSH): https://docs.hpc.uni-mainz.de/docs/advanced/job-monitoring/#accessing-compute-nodes
- HPC Support: hpc@uni-mainz.de
- GitHub: https://github.com/NHR-SW/2026-IO-Performance-Analysis

Thank you!

Website: https://nhrsw.de/

GitHub: https://github.com/NHR-SW/2026-IO-Performance-Analysis

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